AJ SMART BIZ · DOCUMENT 05

Database Design

Conventions, entity relationships and a table-by-table reference for all 52 tables.

Audience	Developers, DBAs, data analysts
Version / date	1.0 · 7 November 2026
Source	`Aj-Smart-Biz-Backend/src/models/*.model.js` and `src/models/index.js` (authoritative). `docs/schema.dbml` is older and misses 13 tables.

Contents

1. Engine and conventions
2. Core entity relationships
3. Platform masters
4. Tenant core & identity
5. Website content
6. Services & bookings
7. Catalogue
8. Orders & customers
9. Inventory
10. Counter invoicing
11. Billing & subscriptions
12. Leads & tracking
13. Platform marketing
14. Delete semantics
15. Schema change process

1. Engine and conventions

Engine	MySQL (utf8mb4, timezone +05:30). SQLite supported for local work (`DB_DIALECT=sqlite`).
Schema management	`sequelize.sync({ alter: true })` on boot (`DB_SYNC_MODE` = alter | force (dev only) | none). There are no migration files.
Primary keys	`id INT UNSIGNED AUTO_INCREMENT` ; `BIGINT` for `lead_visits` and `marketing_lead_visits` .
Naming	Tables snake_case plural, explicit `tableName` ; tenant tables prefixed `company_` . Model attributes camelCase → columns snake_case (`underscored: true`).
Common columns	`status ENUM('active','inactive')` , `created_by` , `updated_by` , `created_at` , `updated_at` , `deleted_at` .
Soft delete	All tables paranoid except `company_functionalities` , `company_invoice_items` , `company_stock` , `company_stock_movements` , `lead_visits` , `marketing_lead_visits` .
Uniqueness	Business keys (company code, slugs, warehouse code…) are checked in services, not by unique indexes, because soft-deleted rows would block reuse. Exceptions: `company_domain.domain` ; composite (company_id, key) on `company_functionalities` (MySQL only).
Branch scoping	Website content tables carry nullable `branch_id` : NULL = company-wide; a branch row overrides it.
JSON columns	Plan features/functionalities, plan snapshot, product images/highlights/specs, service highlights/days, settings blobs, blog tags.
Associations	Declared only in `src/models/index.js` .

2. Core entity relationships

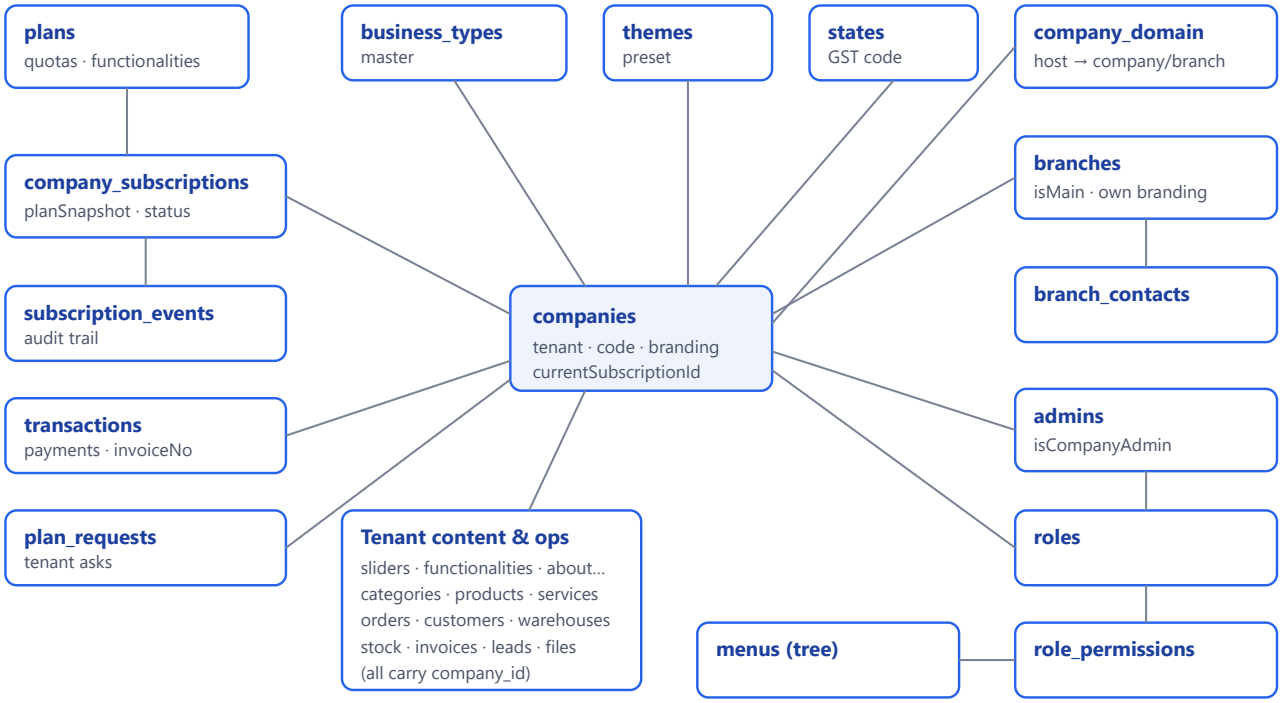


Figure 1 — Core relationships. A company has many of almost everything (CASCADE); history tables use SET NULL so records survive.

3. Platform masters

Table	Purpose	Key columns
states	Indian states (8 seeded)	name, code, country, gst_code
business_types	Trades (7 seeded)	name, slug, icon, description, is_public
themes	Colour presets	name, code, primary/secondary/accent/text/background/sidebar colours, font_family, mode, preview_image, config, is_default
plans	What is sold	name, code, price, discount_price, currency, billing_cycle, duration_days, trial_days, max_branches, max_admins, max_users, storage_mb, features (JSON), functionalities (JSON), is_popular, is_public, show_in_admin, sequence
super_admins	Platform staff	name, email, phone, password (bcrypt, hidden by default scope), role (super_admin|staff), is_root, last_login_at

4. Tenant core & identity

Table	Purpose	Key columns
companies	The tenant	name, code, legal_name, description, business_type_id, theme_id, theme_config, state_id, contact/address, gst, pan, logo, favicon, currency, timezone, storage_limit_mb, current_subscription_id, subscription_end_date
branches	Trading places	company_id, name, code, is_main, logo, favicon, theme_config, contact/address, latitude/longitude, opening/closing time
branch_contacts	People per branch	branch_id, company_id, name, designation, department, email, phone, is_primary, notes
company_domain	Host → tenant	domain (unique, lower-case), company_id, sub_company_id (branch pin), is_primary
roles	Company roles	company_id, name, slug, description, is_system
menus	Console menu tree	company_id (NULL = system), parent_id, name, slug, icon, route, sequence, is_system
role_permissions	RBAC matrix	company_id, role_id, menu_id, can_view, can_create, can_edit, can_delete, can_export
admins	Company users	company_id, branch_id, role_id, name, email, phone, password, avatar, is_company_admin, must_change_password, last_login_at
company_files	Storage ledger	company_id, path, folder, size_bytes, mime_type, uploaded_by

5. Website content

Table	Branch-aware	Key columns
sliders	Yes	title, eyebrow, subtitle, image, mobile_image, cta_label, cta_url, sequence
company_functionalities	—	key (enum of FUNCTIONALITY), settings (JSON), status — one row per (company, key), not paranoid
company_whatsapp_numbers	—	type (inquiry|contact|support|orders), country_code, number, label, default_message, sequence
company_social_links	No	platform, url, label, sequence
company_about	Yes	nav_label, eyebrow, title, lead, body
company_contact	Yes	nav_label, eyebrow, title, lead, show_form, form_target (email|whatsapp), form_note, show_locations

Table	Branch-aware	Key columns
company_stats	Yes	label, value, mode (fixed|since_date), since_date, unit, prefix, suffix
company_team_members	Yes	name, role, bio, photo, email, phone, linkedin_url
company_gallery_items	Yes	image, title, caption, alt_text
company_testimonials	Yes	author_name, author_role, photo, rating (≤5), body, source (admin|visitor), moderation (pending|approved|rejected), moderated_at/by, submitted_ip
company_features	Yes	title, body, icon (FEATURE_ICONS key)
company_blog_posts	—	slug, title, excerpt, body, cover_image, author, tags (JSON), published_at, featured

6. Services & bookings

Table	Key columns
company_service_categories	parent_id, name, slug, description, image, icon, featured, sequence
company_services	category_id (SET NULL), slug, image, icon, title, summary, highlights, price_label, price, offer_price, on_offer, description, duration_minutes, bookable, slot_capacity, open_time, close_time, slot_minutes, days, show_price, cta_label, featured
service_leads	service_id (SET NULL) + service_title copy, name, phone, service_price_label, amount, currency, payment_status, payment_reference, paid_at, customer_id, booking_date, booking_time, booking_minutes, booking_status, booking_response, stage, note, sent_to_whatsapp, source_url, submitted_ip

7. Catalogue

Table	Key columns
company_categories	parent_id (self, max depth 3), name, slug, description, image, icon, featured, sequence
company_products	category_id (SET NULL), name, slug, sku, summary, description, images (JSON ≤ 8), highlights, specs, price, offer_price, price_label, on_offer, offer_label, cta_label, stock_status (in_stock|out_of_stock|made_to_order), track_inventory, featured, orderable

8. Orders & customers

Table	Key columns
company_orders	order_no (ORD#####), mode, customer_id (SET NULL), customer name/phone/address/note copies, item_count, unpriced_items, subtotal, adjustment, adjustment_note, total, currency, status, payment_status, payment_reference, warehouse_id, stock_reserved, stock_issued, sent_to_whatsapp, source_url, submitted_ip, internal_note, cancel_reason, confirmed/packed/dispatched/delivered/cancelled_at
company_order_items	order_id, product_id (SET NULL), product_name, sku, slug, quantity, unit_price, list_price, price_label, line_total
customers	company_id, name, phone (unique per company in service), email, otp_hash, otp_expires_at, otp_attempts, phone_verified, last_login_at, notes
customer_addresses	customer_id, label, contact_name, contact_phone, line1, line2, landmark, city, state_id, pincode, is_default (≤10 per customer)

9. Inventory

Table	Key columns
company_warehouses	branch_id (optional), name, code (unique per company), address, state_id, pincode, contact, is_default (exactly one), notes, sequence
company_stock	warehouse_id, product_id, quantity, reserved, reorder_level, bin_location, last_movement_at — running total
company_stock_movements	warehouse_id, product_id, type (in|out|adjust), reason (opening|purchase|sale|return|damage|correction|transfer|invoice), quantity, balance_after, unit_cost, order_id, invoice_id, counter_warehouse_id, reference, note — append-only ledger

10. Counter invoicing

Table	Key columns
company_invoices	branch_id, warehouse_id, customer_id, invoice_no (INV#####), invoice_date, due_date, customer snapshot incl. GSTIN, item_count, subtotal, item_discount, tax_total, bill_discount_type/value, bill_discount, round_off, grand_total, amount_paid, status (draft|issued|cancelled), payment_status (unpaid|partial|paid), payment_mode, payment_reference, notes, stock_applied, issued_at/by, cancelled_at/by, cancel_reason
company_invoice_items	product snapshot, quantity, unit_price, list_price, gross, discount_type/value, discount, taxable, tax_percent (0/5/12/18/28), tax, line_total, stock_deducted, sequence

11. Billing & subscriptions

Table	Key columns
company_subscriptions	company_id, plan_id, plan_snapshot, start_date, end_date, amount, discount, tax_amount, total_amount, currency, credit_applied, is_trial, auto_renew, grace_days, change_type, status, previous_subscription_id, activated/suspended/cancelled/ended_at, remarks
subscription_events	subscription_id, company_id, type, from/to_status, from/to_plan_id, amount, effective_at, reason, meta, actor_id, actor_name
plan_requests	requested_plan_id, current_plan_id, type, status (pending|approved|rejected|cancelled), note, decision_note, requested_by, decided_by, decided_at, resulting_subscription_id
transactions	subscription_id, plan_id, invoice_no (INV-YYYY-000001), amount, discount, tax_amount, total_amount, currency, payment_mode, payment_reference, paid_at, status, remarks

12. Leads & tracking

Table	Key columns
leads	One per device per company: device_id, name, email, phone, fcm_token, visit/page_view counts, first/last seen and landing, referrer, utm_* columns, device/os/browser, device_type, is_bot, ip, geo, language, stage (new|contacted|qualified|converted|lost), notes
lead_visits	Session (30 min): url, path, title, host, referrer, utm, user_agent, device and screen data, geo/location JSON, extra JSON

13. Platform marketing

Table	Key columns
marketing_contents	key (site, contact, hero, steps, promises, recharge, cta, services_page, plans_page, demo_page, about_page, contact_page, faq_intro, templates_page), label, payload JSON
marketing_faqs	question, answer, sequence
marketing_templates	company_id (demo tenant), title, summary, demo_url, design (default|midnight|atelier), featured
marketing_enquiries	kind (demo|contact), name, business_name, phone, email, city, business_type, plan, message, source, status, note, ip_address, user_agent
marketing_leads / marketing_lead_visits	As leads/lead_visits without company; enquiry_id link (SET NULL)

14. Delete semantics

- **CASCADE** from companies and branches for owned content (sliders, cards, products, stock rows).
- **SET NULL / no constraint** for records of money and history: orders, service leads, order items and stock movements referencing products, invoice pointers.
- **Snapshots** copied onto orders, order items, invoices, service leads and subscriptions so history survives edits and deletes.
- Deleting a category promotes subcategories and un-categorises products in one transaction (done by hand because a soft delete never fires `ON DELETE SET NULL`).

15. Schema change process

1. Edit or add `src/models/<name>.model.js`; register and associate in `src/models/index.js`.
2. Restart the API: alter-sync adds tables and columns. Backfills go in an idempotent `ensureX()` step in `src/seeders/bootstrap.js`.
3. Back up production before deploying a model change; alter-sync can rebuild tables on SQLite and must never run with `force` in production (boot refuses it).
4. Update `docs/schema.dbml` and this document.